

DNA-omic Feature Selection

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- 1 Analyze and visualize the relationship between TMB (Tumor Mutation Burden) and pCR (pathological complete response) in breast cancer samples, and investigate the influence of HER2 status on this relationship. 9

```
library(broom)
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(readr)
library(stringr)
library(tidyr)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.0      v purrr 1.0.1
## v ggplot2 3.5.2      v tibble 3.2.1
## v lubridate 1.9.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tibble)
library(ggplot2)
library(ggpubr)
library(ggsignif)
```

```

library(ggrepel)
# Read mutation and clinical data
mut_df <- read_excel("41586_2021_4278_MOESM4_ESM.xlsx", sheet = 2)
clin_df <- read_excel("41586_2021_4278_MOESM4_ESM.xlsx", sheet = 1)

# Calculate mutation frequency of each gene across patients
mut_counts <- mut_df %>%
  select(Donor.ID, Hugo_Symbol) %>%
  distinct() %>%
  group_by(Hugo_Symbol) %>%
  summarise(mutated_samples = n()) %>%
  mutate(freq = mutated_samples / n_distinct(mut_df$Donor.ID)) %>%
  filter(freq > 0.05) # Keep genes mutated in >5% of samples

selected_genes <- mut_counts$Hugo_Symbol

# Create a wide-format matrix indicating whether each gene is mutated in each patient
mut_matrix <- mut_df %>%
  filter(Hugo_Symbol %in% selected_genes) %>%
  select(Donor.ID, Hugo_Symbol) %>%
  distinct() %>%
  mutate(mutated = 1) %>%
  pivot_wider(names_from = Hugo_Symbol, values_from = mutated, values_fill = 0)

# Merge with clinical data
merged_df <- clin_df %>%
  select(Donor.ID, pCR.RD) %>%
  mutate(pCR = ifelse(pCR.RD == "pCR", 1, 0)) %>%
  left_join(mut_matrix, by = "Donor.ID") %>%
  replace(is.na(.), 0)

# Identify gene columns
gene_cols <- setdiff(colnames(merged_df), c("Donor.ID", "pCR.RD", "pCR"))

# Run univariable logistic regression: pCR ~ each gene
results_list <- lapply(gene_cols, function(gene) {
  model <- glm(pCR ~ merged_df[[gene]], data = merged_df, family = binomial)
  est <- summary(model)$coefficients[2, ]
  ci <- suppressMessages(confint(model))[2, ]
  data.frame(
    Gene = gene,
    OR = exp(est["Estimate"]),
    CI_lower = exp(ci[1]),
    CI_upper = exp(ci[2]),
    p_value = est["Pr(>|z|)"]
  )
})

```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

[illegible]

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
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## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
results_df <- bind_rows(results_list)
```

```
# Add labels and color categories
```

```
results_df <- results_df %>%
  mutate(
    label = paste0("OR = ", round(OR, 2),
                  "\nCI: [", round(CI_lower, 2), ", ", round(CI_upper, 2), "]",
                  "\nP = ", signif(p_value, 2)),
    color = case_when(
      p_value > 0.05 ~ "nonsignificant",
      OR >= 1 ~ "risk",
      OR < 1 ~ "protective"
    )
  )
```

```
# Paginate (10 genes per page)
```

```
genes_per_page <- 10
results_df <- results_df %>%
  arrange(desc(OR)) %>%
  mutate(page = ceiling(row_number() / genes_per_page))
```

```
n_pages <- max(results_df$page)
```

```
# Generate paginated forest plots
```

```
plots <- lapply(1:n_pages, function(p) {
  ggplot(results_df %>% filter(page == p), aes(x = reorder(Gene, OR), y = OR)) +
    geom_point(aes(color = color), size = 4) +
    geom_errorbar(aes(ymin = CI_lower, ymax = CI_upper, color = color), width = 0.2, size = 1) +
    geom_hline(yintercept = 1, linetype = "dashed", color = "gray40") +
    coord_flip() +
    scale_color_manual(values = c("risk" = "red", "protective" = "blue", "nonsignificant" = "gray60")) +
    scale_y_log10() +
    geom_text(aes(label = label), hjust = -0.1, size = 4) +
```

```

theme_minimal(base_size = 14) +
labs(
  y = "Odds Ratio (log scale)",
  x = NULL,
  title = paste("Forest Plot - Page", p)
)
})

```

```

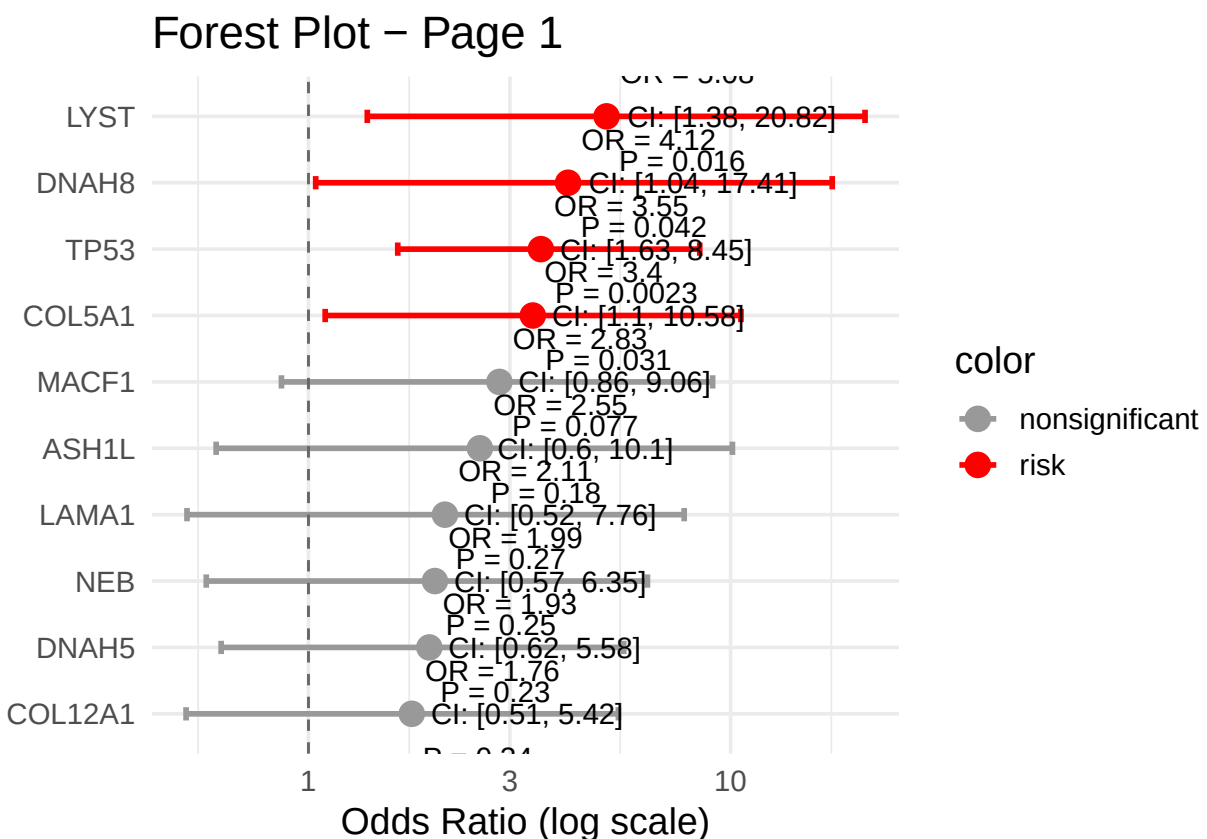
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

```

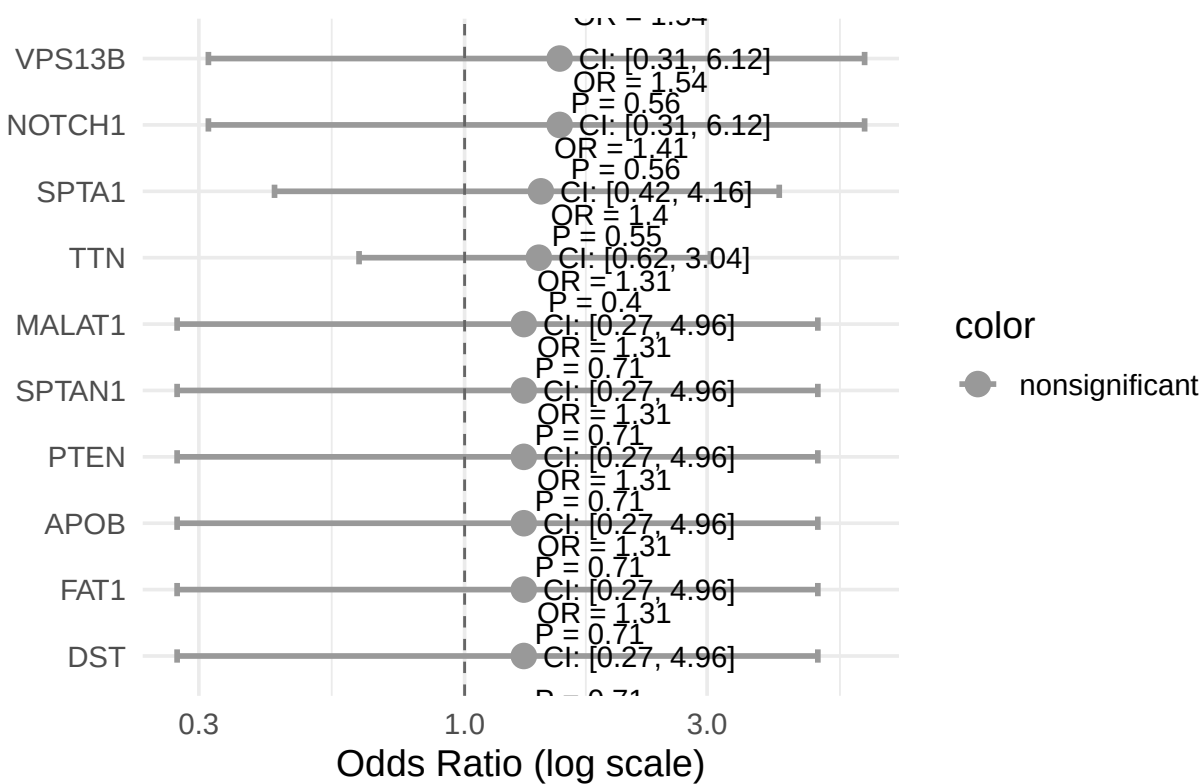
# Print each page and insert page break
for (p in 1:n_pages) {
  print(plots[[p]])
  cat("\n\\newpage\n") # For PDF output, forces a new page
}

```



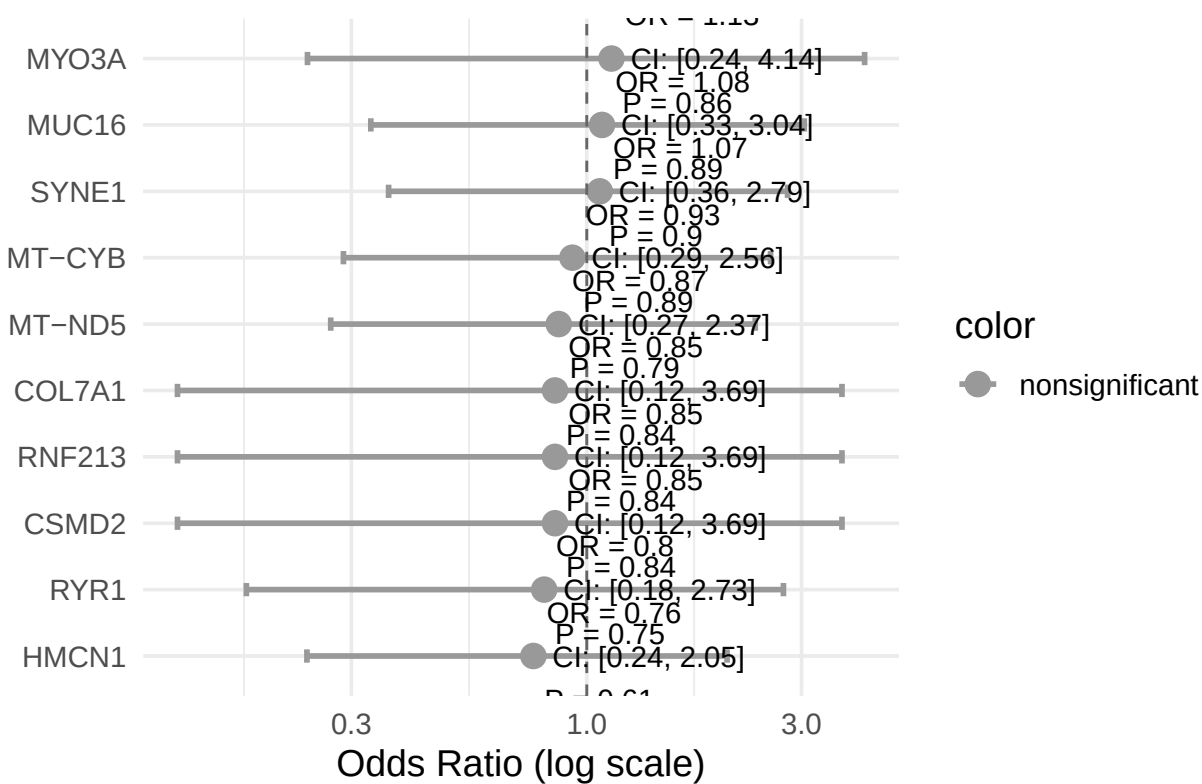
```
## \newpage
```

Forest Plot – Page 2



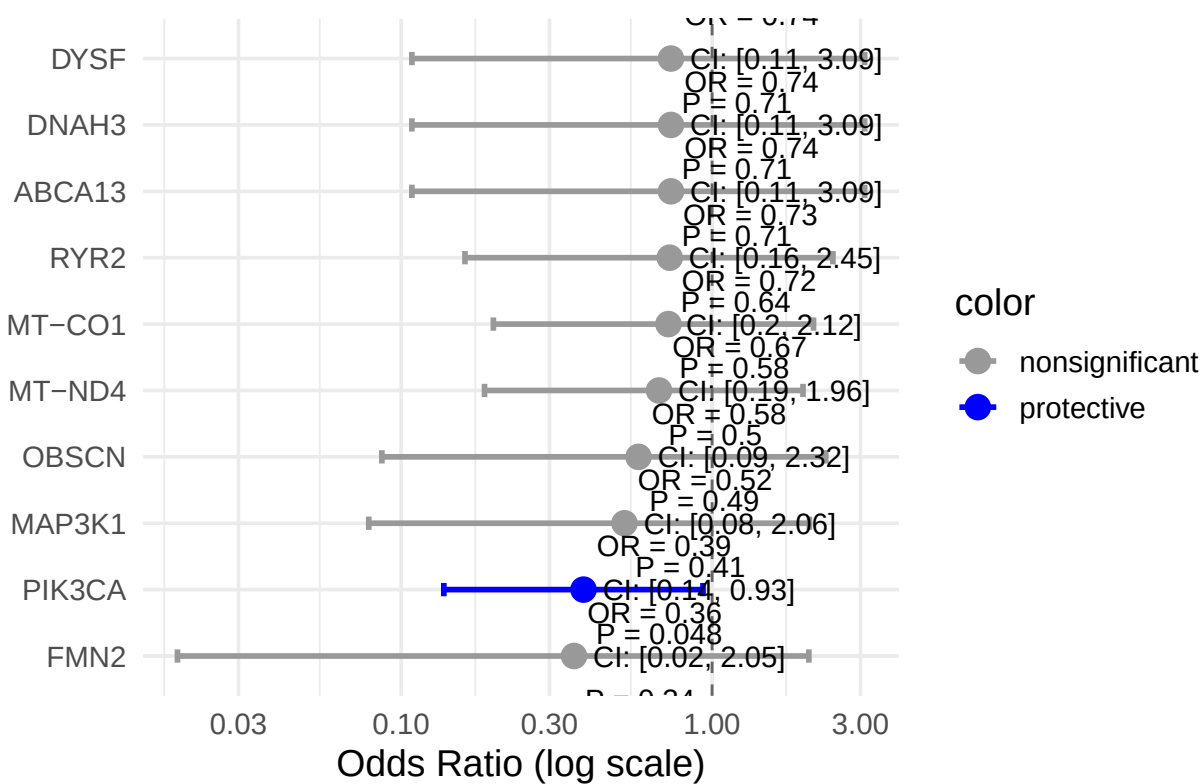
\newpage

Forest Plot – Page 3



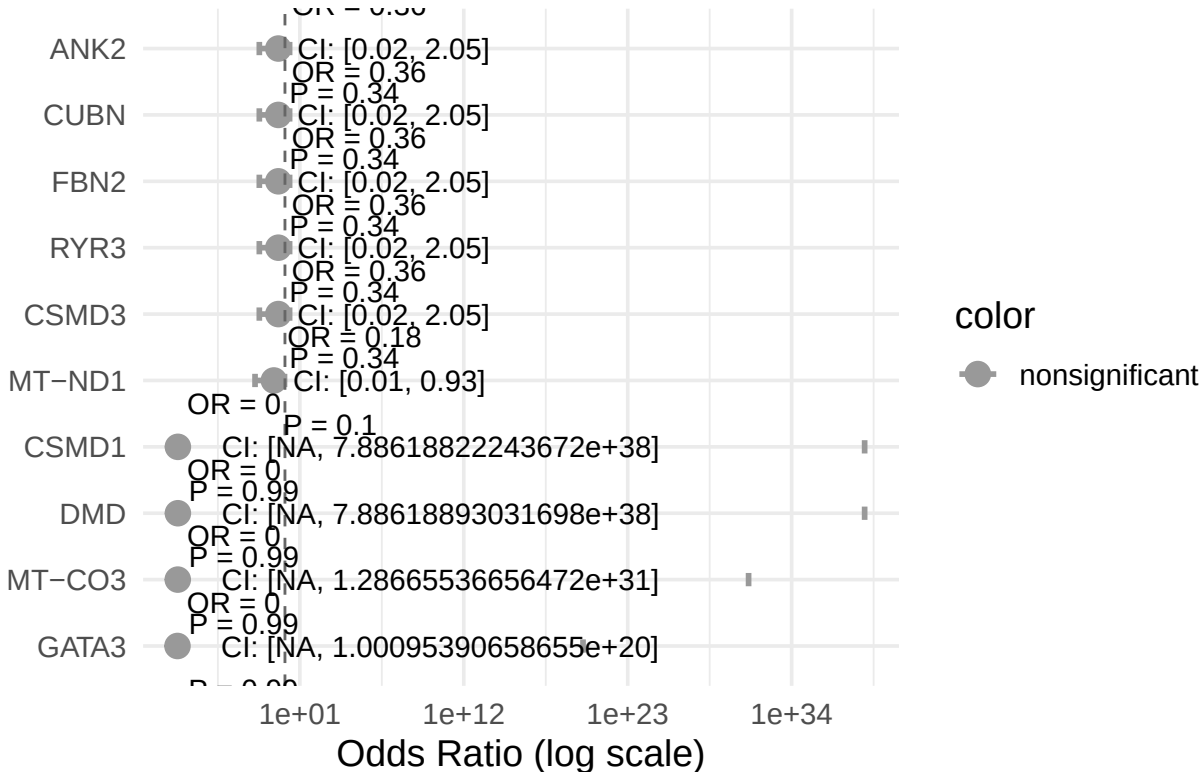
\newpage

Forest Plot – Page 4



\newpage

Forest Plot – Page 5



\newpage

Use logistic regression to judge if the genes have effect on pCR.

According to the forest plot, TP53 mutation was significantly associated with a higher likelihood of achieving pCR, represented in red, indicating that it may have a positive affect on therapy response.

PIK3CA mutation was also significantly associated with pCR, but in a negative way.

The genes with gray color were not significantly have relationship the pCR.

1 Analyze and visualize the relationship between TMB (Tumor Mutation Burden) and pCR (pathological complete response) in breast cancer samples, and investigate the influence of HER2 status on this relationship.

```
#define coding TMB
coding_types <- c("Missense_Mutation", "Nonsense_Mutation", "Silent",
                  "Splice_Site", "In_Frame_Del", "In_Frame_Ins",
                  "Frame_Shift_Del", "Frame_Shift_Ins")

# Mutation number per sample (can be used as a simplified version of TMB)
tmb_df <- mut_df %>%
  filter(MAF_Variant %in% coding_types) %>%
```

```

group_by(Donor.ID) %>%
  summarise(CodingTMB = n())

head(tmb_df)

```

```

## # A tibble: 6 x 2
##   Donor.ID CodingTMB
##   <chr>      <int>
## 1 T001        118
## 2 T002         94
## 3 T003         59
## 4 T004         80
## 5 T005         78
## 6 T006         19

```

```

# Combine TMB with clinical data
merged_df <- clin_df %>%
  left_join(tmb_df, by = "Donor.ID") %>%
  mutate(
    pCR_binary = case_when(
      pCR.RD == "pCR" ~ "pCR",
      pCR.RD == "RD" ~ "RD",
      TRUE ~ NA_character_
    ),
    HER2.status = toupper(HER2.status),
    HER2_group = case_when(
      HER2.status == "POS" ~ "HER2+",
      HER2.status == "NEG" ~ "HER2-",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(pCR_binary)) %>% # ← NA
  left_join(mut_df, by = "Donor.ID")
# Overall comparison of TMB (median + test)
library(ggpubr)

# View the median
merged_df %>%
  group_by(pCR_binary) %>%
  summarise(median_CodingTMB = median(CodingTMB, na.rm = TRUE))

```

```

## # A tibble: 2 x 2
##   pCR_binary median_CodingTMB
##   <chr>      <dbl>
## 1 RD          78
## 2 pCR        108

```

```

# Wilcoxon test
wilcox.test(CodingTMB ~ pCR_binary, data = merged_df)

```

```

##
## Wilcoxon rank sum test with continuity correction

```

```
##
## data: CodingTMB by pCR_binary
## W = 31575196, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

```
# HER2- group
wilcox.test(CodingTMB ~ pCR_binary, data = merged_df %>% filter(HER2_group == "HER2-"))
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: CodingTMB by pCR_binary
## W = 11884282, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
```

```
# HER2+ group
wilcox.test(CodingTMB ~ pCR_binary, data = merged_df %>% filter(HER2_group == "HER2+"))
```

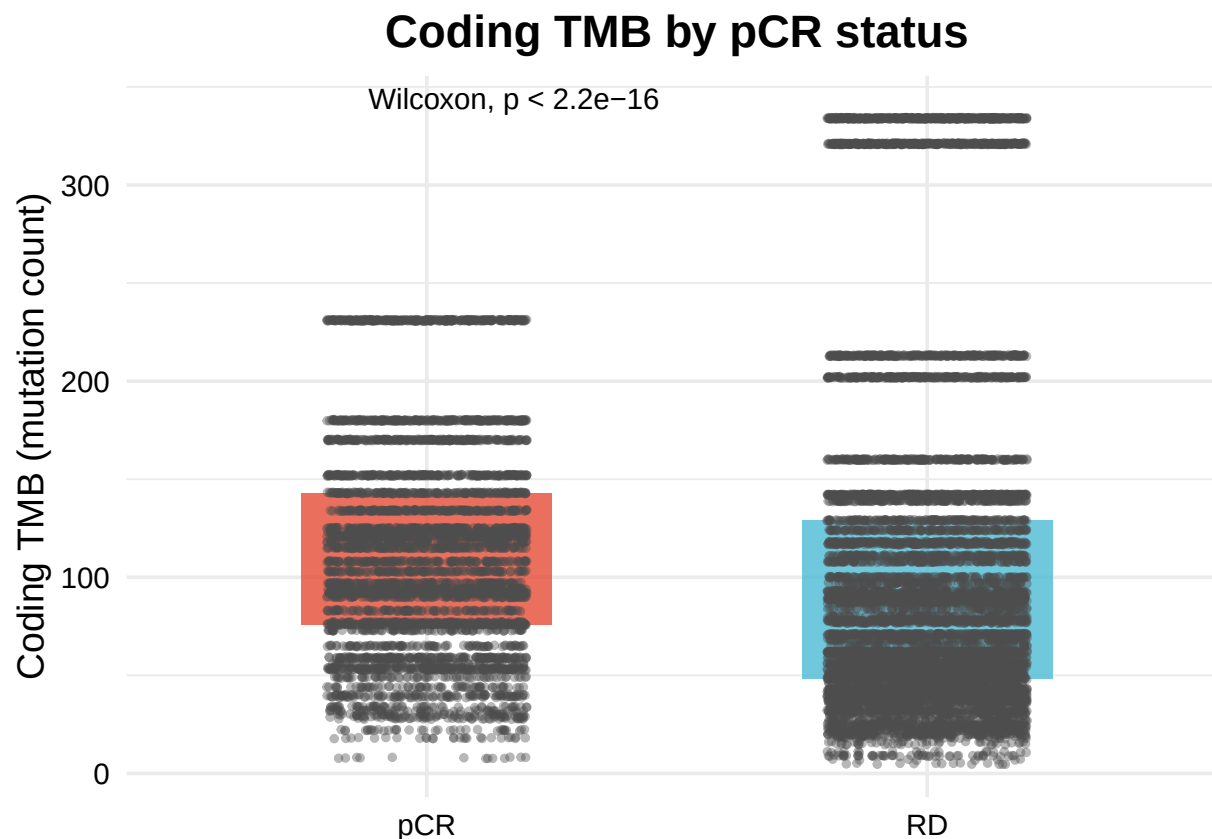
```
##
## Wilcoxon rank sum test with continuity correction
##
## data: CodingTMB by pCR_binary
## W = 4682423, p-value = 1.71e-08
## alternative hypothesis: true location shift is not equal to 0
```

```
# Drawing
ggplot(merged_df, aes(x = pCR_binary, y = CodingTMB, fill = pCR_binary)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.8, width = 0.5, color = NA) +
  geom_jitter(width = 0.2, alpha = 0.4, color = "gray30", size = 1) +
  stat_compare_means(method = "wilcox.test", label.y = max(merged_df$CodingTMB, na.rm = TRUE) + 5) +
  labs(title = "Coding TMB by pCR status",
       y = "Coding TMB (mutation count)", x = NULL) +
  scale_fill_manual(values = c("pCR" = "#E64B35", "RD" = "#4DBBD5")) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none",
       plot.title = element_text(hjust = 0.5, face = "bold"),
       axis.text = element_text(color = "black"))
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_compare_means()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



Comparison of coding tumor mutation burden (TMB) between pCR and RD groups. Each dot represents one patient. Boxplots show the distribution of coding TMB across the two groups, with a Wilcoxon rank-sum test indicating a statistically significant difference ($p < 2.2e-16$). The pCR group had higher coding TMB compared to the RD group.

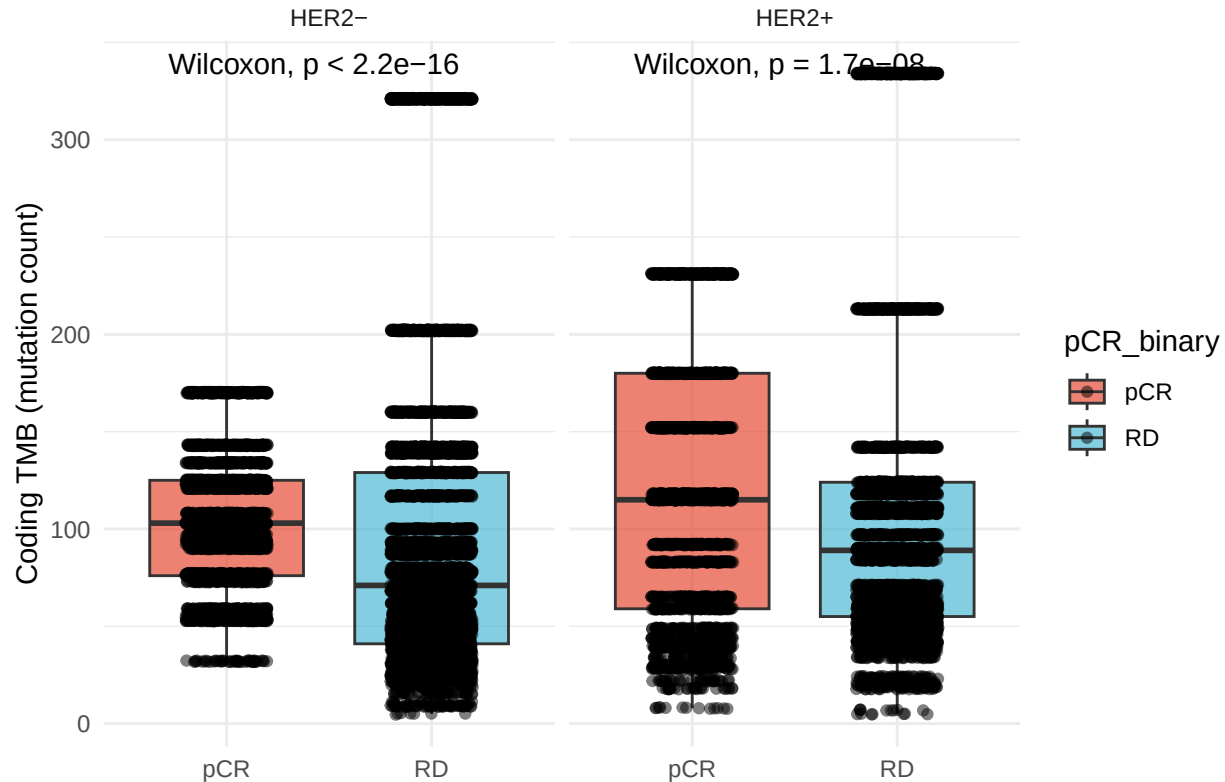
```
ggplot(merged_df, aes(x = pCR_binary, y = CodingTMB, fill = pCR_binary)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.7) +
  geom_jitter(width = 0.2, alpha = 0.5, color = "black") +
  stat_compare_means(method = "wilcox.test") +
  facet_wrap(~ HER2_group) +
  labs(title = "Coding TMB by pCR and HER2 status", y = "Coding TMB (mutation count)", x = NULL) +
  theme_minimal() +
  scale_fill_manual(values = c("pCR" = "#E64B35", "RD" = "#4DBBD5"))
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_compare_means()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

Coding TMB by pCR and HER2 status



Comparison of coding tumor mutation burden (TMB) between pCR and RD groups stratified by HER2 status. Within both HER2- and HER2+ subgroups, patients who achieved pCR showed significantly higher TMB compared to those with residual disease (RD). Wilcoxon rank-sum test: $p < 2.2e-16$ (HER2-), $p = 1.7e-08$ (HER2+).

#To investigate whether the HRD (Homologous Recombination Deficiency) score is related to the neoadjuvant therapy response (pCR), and to consider the stratified effect of HER2 status at the same time.

```
# Import HRD data
hrd_df <- read_tsv("transneo-diagnosis-HRD.tsv")

## Rows: 168 Columns: 5
## -- Column specification -----
## Delimiter: "\t"
## chr (1): Trial.ID
## dbl (4): HRD, Telomeric.AI, LST, HRD.sum
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

# Create pCR binarization variables & HER2 groups (HER2- / HER2+)
clin_df <- clin_df %>%
  mutate(
    pCR_binary = ifelse(pCR.RD == "pCR", 1, 0),
    HER2_group = ifelse(HER2.status == "POS", "HER2+", "HER2-")
  )
```

```
# Combine HRD with clinical data
merged <- clin_df %>%
  inner_join(hrd_df, by = c("Donor.ID" = "Trial.ID"))

# Association analysis of HRD and pCR in the whole cohort: Logistic regression
model_hrd <- glm(pCR_binary ~ HRD.sum, data = merged, family = binomial)
summary(model_hrd)
```

```
##
## Call:
## glm(formula = pCR_binary ~ HRD.sum, family = binomial, data = merged)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3496  -0.7406  -0.6058  -0.1609   1.9613
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.25476     0.38659  -5.832 5.46e-09 ***
## HRD.sum      0.04078     0.01133   3.600 0.000319 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 188.94  on 167  degrees of freedom
## Residual deviance: 174.90  on 166  degrees of freedom
## AIC: 178.9
##
## Number of Fisher Scoring iterations: 3
```

```
exp(cbind(OR = coef(model_hrd), confint(model_hrd))) # OR + CI
```

```
## Waiting for profiling to be done...
```

```
##              OR      2.5 %    97.5 %
## (Intercept) 0.1048987 0.04719719 0.2164179
## HRD.sum     1.0416252 1.01939976 1.0660526
```

```
# HRD.sum non-parametric test in pCR vs non-PCR
wilcox.test(HRD.sum ~ pCR_binary, data = merged)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data:  HRD.sum by pCR_binary
## W = 1669.5, p-value = 0.0003483
## alternative hypothesis: true location shift is not equal to 0
```

```
# HER2 subtype stratified analysis
```

```
# HER2- group
```

```
wilcox.test(HRD.sum ~ pCR_binary, data = merged %>% filter(HER2_group == "HER2-"))
```

```
##
```

```
## Wilcoxon rank sum test with continuity correction
```

```
##
```

```
## data: HRD.sum by pCR_binary
```

```
## W = 310, p-value = 2.973e-06
```

```
## alternative hypothesis: true location shift is not equal to 0
```

```
# HER2+ group
```

```
wilcox.test(HRD.sum ~ pCR_binary, data = merged %>% filter(HER2_group == "HER2+"))
```

```
## Warning in wilcox.test.default(x = DATA[[1L]], y = DATA[[2L]], ...): cannot
```

```
## compute exact p-value with ties
```

```
##
```

```
## Wilcoxon rank sum test with continuity correction
```

```
##
```

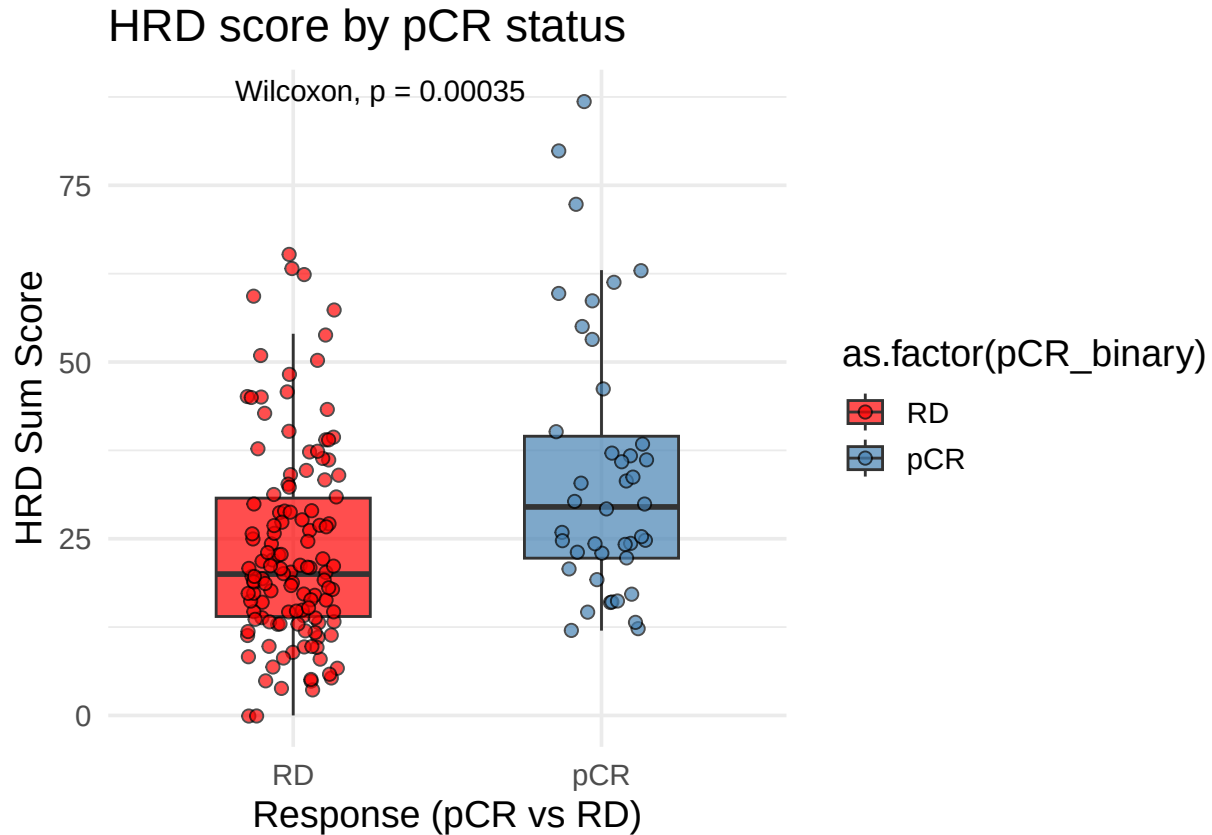
```
## data: HRD.sum by pCR_binary
```

```
## W = 483.5, p-value = 0.6387
```

```
## alternative hypothesis: true location shift is not equal to 0
```

```
# Visualization of HRD vs pCR (full cohort)
```

```
ggplot(merged, aes(x = as.factor(pCR_binary), y = HRD.sum, fill = as.factor(pCR_binary))) +  
  geom_boxplot(alpha = 0.7, outlier.shape = NA, width = 0.5) +  
  geom_jitter(width = 0.15, shape = 21, size = 2, alpha = 0.7, color = "black") +  
  stat_compare_means(method = "wilcox.test") +  
  scale_fill_manual(values = c("red", "steelblue"), labels = c("RD", "pCR")) +  
  scale_x_discrete(labels = c("RD", "pCR")) +  
  labs(  
    x = "Response (pCR vs RD)",  
    y = "HRD Sum Score",  
    title = "HRD score by pCR status"  
  ) +  
  theme_minimal(base_size = 14)
```



The total HRD score of the patients in the pCR group was significantly higher than that in the RD group (with a higher median and a more right-skewed distribution).

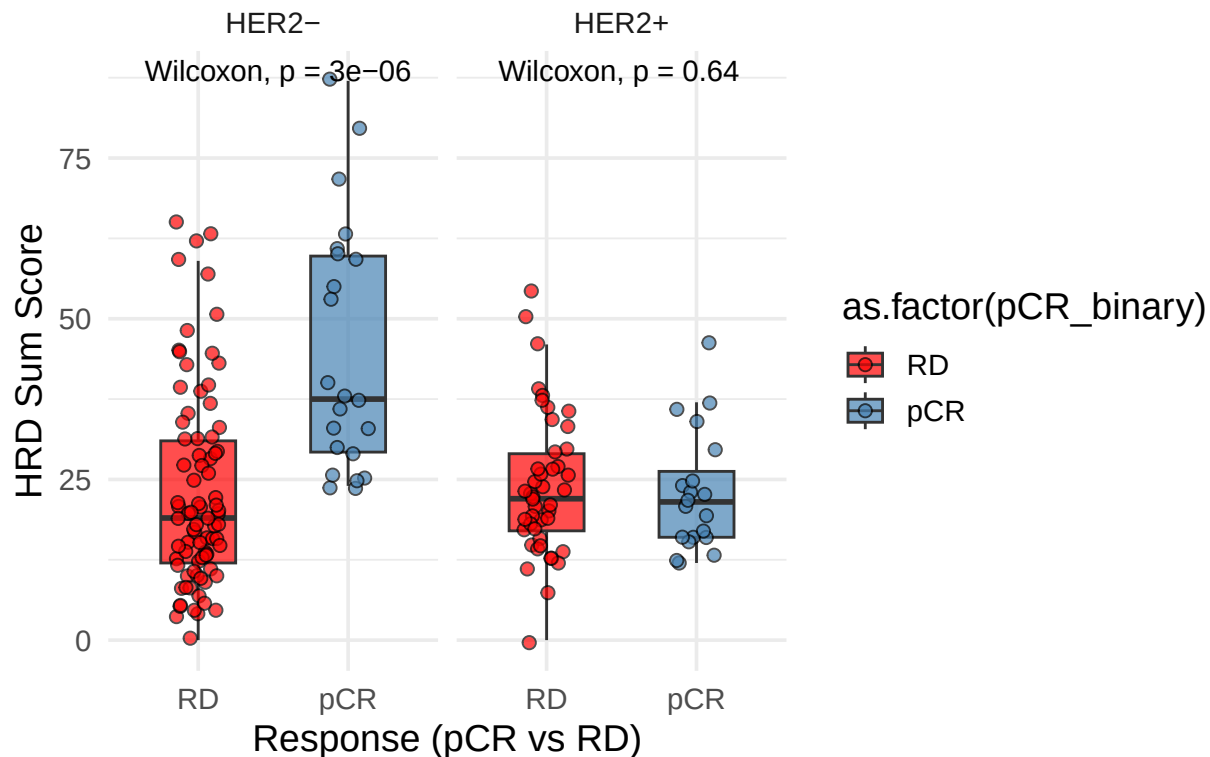
A high HRD score may be associated with a better response to neoadjuvant therapy (that is, it is easier to achieve pCR).

The p value of 0.00035 in the Wilcoxon test indicates a highly significant difference.

Compared to RD patients, those who achieved pCR exhibited significantly higher HRD scores (Wilcoxon $p = 0.00035$), suggesting that increased genomic instability may be associated with better neoadjuvant treatment response.

```
ggplot(merged, aes(x = as.factor(pCR_binary), y = HRD.sum, fill = as.factor(pCR_binary))) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA, width = 0.5) +
  geom_jitter(width = 0.15, shape = 21, size = 2, alpha = 0.7, color = "black") +
  stat_compare_means(method = "wilcox.test") +
  facet_wrap(~ HER2_group) +
  scale_fill_manual(values = c("red", "steelblue"), labels = c("RD", "pCR")) +
  scale_x_discrete(labels = c("RD", "pCR")) +
  labs(
    x = "Response (pCR vs RD)",
    y = "HRD Sum Score",
    title = "HRD score by pCR status and HER2 group"
  ) +
  theme_minimal(base_size = 14)
```

HRD score by pCR status and HER2 group



In the overall cohort, higher HRD scores were significantly associated with pCR ($p = 0.00035$). Stratified analysis revealed that this association was driven by HER2- patients ($p = 3e-06$), whereas no significant difference was observed in the HER2+ subgroup ($p = 0.64$), suggesting a subtype-specific role of HRD in predicting response to neoadjuvant therapy.

#To analyze the relationship between CIN (Chromosomal Instability) and neoadjuvant therapy response (RCB classification and pCR)

```
# read CIN data
cin_df <- read_tsv("transneo-diagnosis-ASCAT-CIN.tsv")

## Rows: 166 Columns: 2
## -- Column specification -----
## Delimiter: "\t"
## chr (1): Trial.ID
## dbl (1): cin
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

colnames(cin_df) <- c("Trial.ID", "CIN")

# Data cleaning and merging
merged_df <- clin_df %>%
  rename(Trial.ID = Donor.ID) %>%
```

```

inner_join(cin_df, by = "Trial.ID") %>%
mutate(
  pCR_binary = ifelse(pCR.RD == "pCR", 1, 0),
  RCB_class = factor(RCB.category, levels = c("RCB-I", "RCB-II", "RCB-III", "pCR"))
)

# Test whether CIN is correlated with the monotony of the RCB class
kruskal_test <- kruskal.test(CIN ~ RCB.category, data = merged_df)
print(kruskal_test)

##
## Kruskal-Wallis rank sum test
##
## data: CIN by RCB.category
## Kruskal-Wallis chi-squared = 17.968, df = 4, p-value = 0.001252

# Compare the differences in CIN between pCR and RD
wilcox_test <- wilcox.test(CIN ~ pCR_binary, data = merged_df)
print(wilcox_test)

##
## Wilcoxon rank sum test with continuity correction
##
## data: CIN by pCR_binary
## W = 1583, p-value = 0.0001503
## alternative hypothesis: true location shift is not equal to 0

# Visualize CIN and RCB classes
p1 <- ggplot(merged_df, aes(x = RCB.category, y = CIN)) +
  geom_boxplot(aes(fill = RCB.category), outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.5) +
  scale_fill_brewer(palette = "Set2") +
  theme_minimal() +
  labs(title = "CIN vs RCB category", x = "RCB category", y = "CIN") +
  stat_compare_means(method = "kruskal.test", label.y = max(merged_df$CIN) * 1.05)

# Visualize the distribution of CIN in pCR vs RD
p2 <- ggplot(merged_df, aes(x = factor(pCR_binary, labels = c("RD", "pCR")), y = CIN)) +
  geom_boxplot(aes(fill = factor(pCR_binary)), outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.5) +
  scale_fill_manual(values = c("red", "skyblue")) +
  theme_minimal() +
  labs(title = "CIN vs pCR status", x = "pCR status", y = "CIN") +
  stat_compare_means(method = "wilcox.test", label.y = max(merged_df$CIN) * 1.05)
# Display graphics
ggarrange(p1, p2, ncol = 2, labels = c("A", "B"))

```

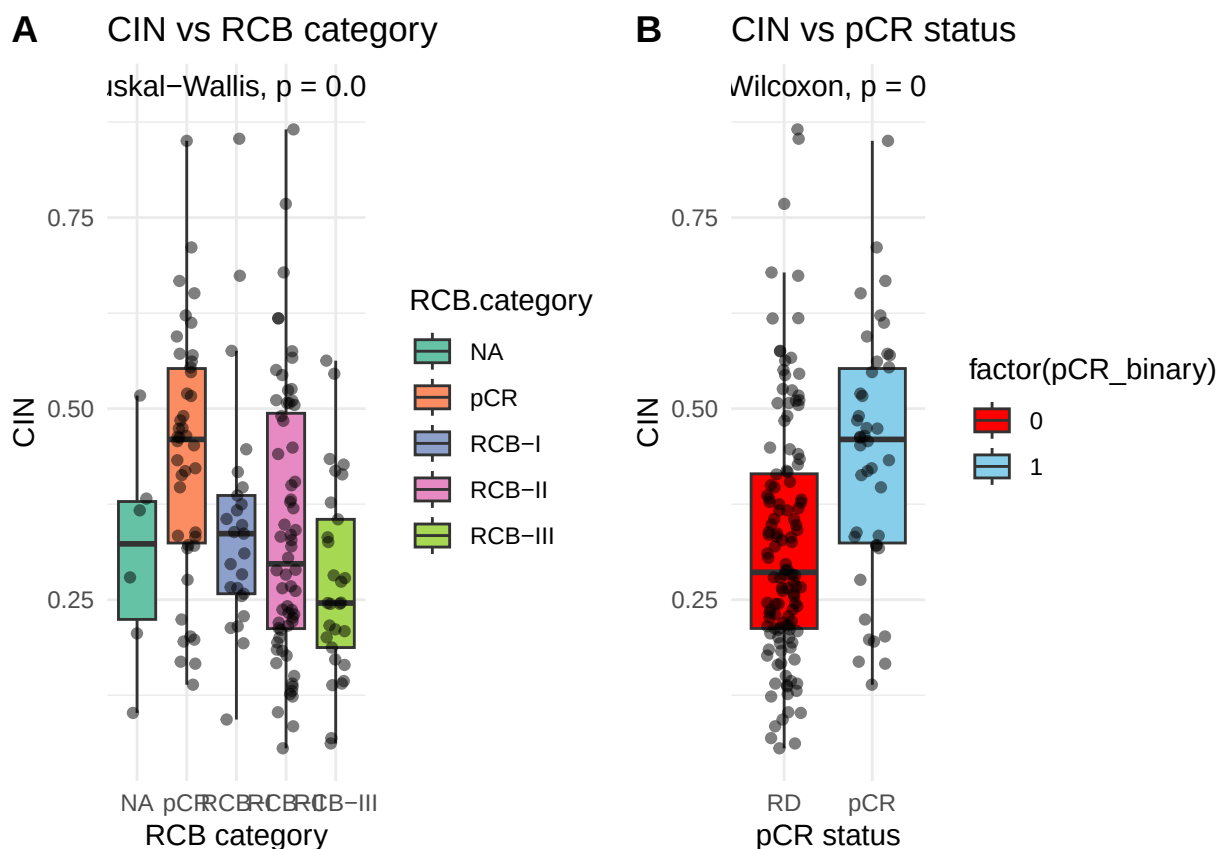


Figure A shows the distribution of ** genomic instability levels (CIN) ** in patients with different RCB categories (including pCR, RCB-I, RCB-II, and RCB-III). It can be seen from the figure that the median of CIN in the pCR group was significantly higher than that in other groups, especially when compared with RCB-I and RCB-II, this difference was more obvious. The Kruskal-Wallis test results indicated that this difference was statistically significant ($p < 0.05$), suggesting that there were systematic differences in CIN levels among different grades of RCB.

Figure B further compared CIN between the pCR and RD groups. It can be seen that the overall CIN level in the pCR group was significantly higher than that in the RD group, and this difference was statistically very significant (Wilcoxon test, $p < 0.001$). It can also be intuitively seen from the box plot that not only is the median of the pCR group higher, but the overall data distribution is also upward.

CIN (Chromosomal Instability) represents the degree of abnormality in the genomic structure of tumors. Higher CIN may lead to more expression of neoantigens or enhance the sensitivity of tumors to drugs, thereby promoting complete response (pCR) of tumor cells to neoadjuvant therapy. Tumors with a lower CIN may have a more stable genomic structure and be more difficult to be destroyed by chemotherapy drugs, and therefore are more likely to present as RD.

Therefore, CIN may be an important biomarker for predicting the response of breast cancer patients to neoadjuvant therapy. In the future, it can be considered to incorporate CIN into the molecular typing model for guiding individualized treatment strategies.

#To investigate whether the iC10 molecular subtype (breast cancer classification) is related to the neoadjuvant therapy response (pCR)

```
# cleaning data
clin_clean <- clin_df %>%
  mutate(
    iC10 = as.factor(iC10),
```

```

    pCR_binary = ifelse(pCR.RD == "pCR", 1, 0) # 1 represents pCR 0 represents RD
  )

# Descriptive statistics: pCR rates were calculated by iC10 grouping
pcr_summary <- clin_clean %>%
  group_by(iC10) %>%
  summarise(
    n = n(),
    pCR_count = sum(pCR_binary, na.rm = TRUE),
    pCR_rate = round(mean(pCR_binary, na.rm = TRUE), 2)
  ) %>%
  arrange(desc(pCR_rate))

print(pcr_summary)

```

```

## # A tibble: 10 x 4
##   iC10      n pCR_count pCR_rate
##   <fct> <int>     <dbl>    <dbl>
## 1 10      27         16     0.59
## 2 5       41         16     0.39
## 3 9       19          5     0.26
## 4 4       13          3     0.23
## 5 1       11          1     0.09
## 6 2       11          1     0.09
## 7 3        9          0      0
## 8 6       15          0      0
## 9 7        8          0      0
## 10 8      14          0      0

```

```

# Chi-square test: Whether iC10 and pCR.RD are independent
table_ic_pcr <- table(clin_clean$iC10, clin_clean$pCR.RD)
chisq_result <- chisq.test(table_ic_pcr)

```

```

## Warning in chisq.test(table_ic_pcr): Chi-squared approximation may be incorrect

```

```

print(chisq_result)

```

```

##
## Pearson's Chi-squared test
##
## data:  table_ic_pcr
## X-squared = 50.644, df = 18, p-value = 6.028e-05

```

```

# Logistic regression model: pCR ~ iC10 (iC1 for the reference group)
model <- glm(pCR_binary ~ iC10, data = clin_clean, family = binomial)
summary(model)

```

```

##
## Call:
## glm(formula = pCR_binary ~ iC10, family = binomial, data = clin_clean)
##

```

```
## Deviance Residuals:
##      Min        1Q      Median        3Q        Max
## -1.34011  -0.78151  -0.00013   0.25565   2.18993
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.303e+00  1.049e+00  -2.195  0.0281 *
## iC102        3.269e-15  1.483e+00   0.000  1.0000
## iC103       -1.626e+01  2.174e+03  -0.007  0.9940
## iC104        1.099e+00  1.238e+00   0.887  0.3750
## iC105        1.856e+00  1.097e+00   1.693  0.0905 .
## iC106       -1.626e+01  1.684e+03  -0.010  0.9923
## iC107       -1.626e+01  2.306e+03  -0.007  0.9944
## iC108       -1.626e+01  1.743e+03  -0.009  0.9926
## iC109        1.273e+00  1.171e+00   1.087  0.2770
## iC1010       2.677e+00  1.120e+00   2.391  0.0168 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 188.94  on 167  degrees of freedom
## Residual deviance: 140.69  on 158  degrees of freedom
## AIC: 160.69
##
## Number of Fisher Scoring iterations: 17

# Show the OR (Odds Ratio) and confidence interval of the regression results
exp_coef <- broom::tidy(model, exponentiate = TRUE, conf.int = TRUE)

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
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```



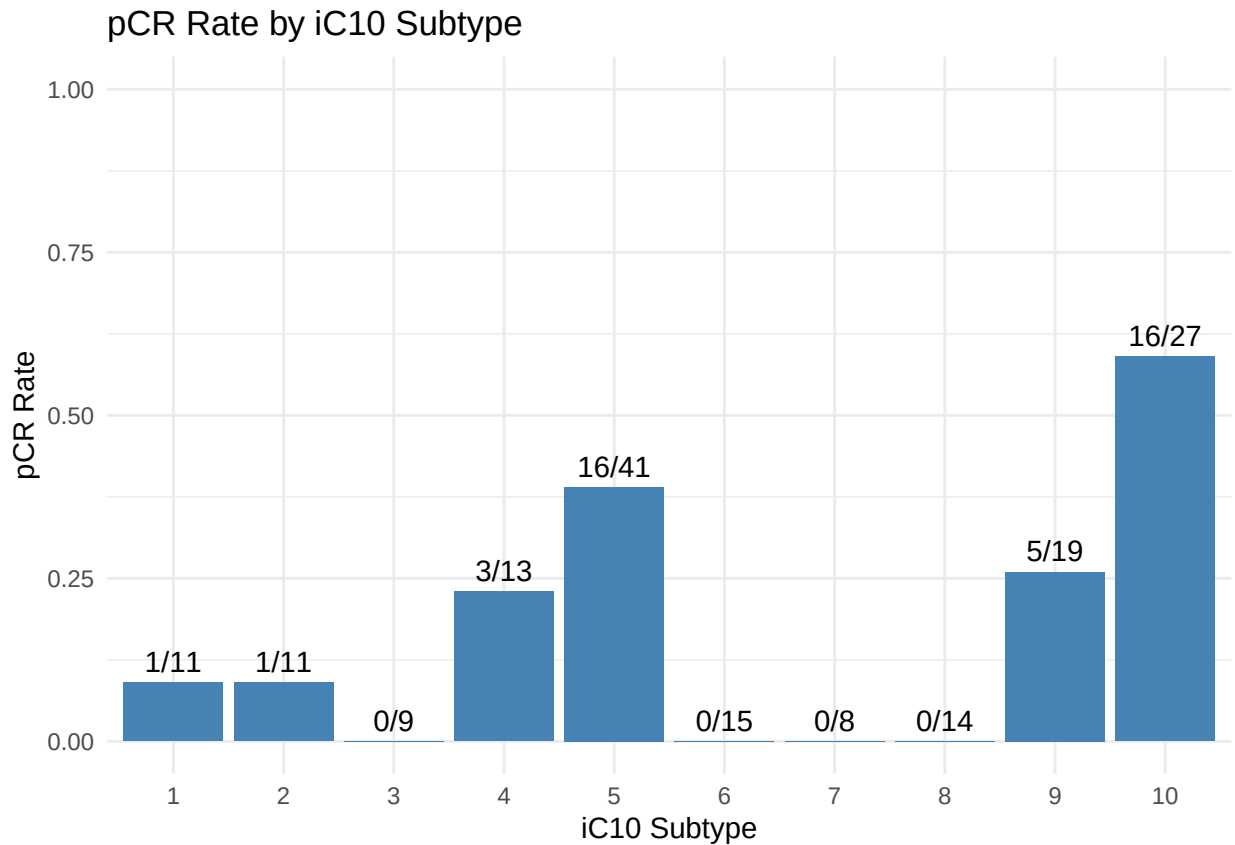
```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
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## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```
print(exp_coef)
```

```
## # A tibble: 10 x 7
##   term                estimate std.error statistic p.value  conf.low conf.high
##   <chr>                <dbl>    <dbl>    <dbl>   <dbl>    <dbl>    <dbl>
## 1 (Intercept)    0.100          1.05 -2.20e+ 0  0.0281  5.45e- 3  5.22e- 1
## 2 iC102           1.00           1.48  2.20e-15  1.00    3.61e- 2  2.77e+ 1
## 3 iC103    0.0000000865    2174.   -7.48e- 3  0.994   4.54e-316  6.31e+33
## 4 iC104           3.00           1.24  8.87e- 1  0.375   3.20e- 1  6.66e+ 1
## 5 iC105           6.40           1.10  1.69e+ 0  0.0905  1.07e+ 0  1.23e+ 2
## 6 iC106    0.0000000865    1684.   -9.66e- 3  0.992   1.39e-246  3.88e+24
## 7 iC107    0.0000000865    2306.   -7.05e- 3  0.994   2.67e-314  1.03e+40
## 8 iC108    0.0000000865    1743.   -9.33e- 3  0.993   1.72e-239  3.34e+28
## 9 iC109           3.57           1.17  1.09e+ 0  0.277   4.75e- 1  7.41e+ 1
## 10 iC1010      14.5           1.12  2.39e+ 0  0.0168  2.30e+ 0  2.87e+ 2
```

```
# Visualization: Bar chart of pCR rates for different iC10 subtypes
ggplot(pcr_summary, aes(x = iC10, y = pCR_rate)) +
  geom_col(fill = "steelblue") +
  geom_text(aes(label = paste0(pCR_count, "/", n)), vjust = -0.5) +
  ylim(0, 1) +
  labs(
    title = "pCR Rate by iC10 Subtype",
```

```
x = "iC10 Subtype",
y = "pCR Rate"
) +
theme_minimal()
```



The pathological complete response (pCR) rates varied substantially across iC10 molecular subtypes. The highest pCR rate was observed in subtype 10 (59.3%), followed by subtype 5 (39.0%) and subtype 9 (26.3%). In contrast, subtypes 3, 6, 7, and 8 showed no pCR cases, indicating potentially poor response to neoadjuvant therapy in these groups. Subtypes 1 and 2 also exhibited low pCR rates (9.1%). These findings suggest that iC10 subtypes may be associated with differential therapeutic response.